

Disaster Recovery Step by Step

- Immediate hands-on application
- Learn step-by-step DR implementation
- Risk, impact, plan, test, improve
- Use templates and real examples
- Realistic CloudCore use case
- Practical tools included
- Build your company’s DR plan
- Tailored for all experience levels

Course Objectives and Learning Approach

- Learn DR fundamentals and terminology
- Design recovery strategies and documentation
- Ensure continuity and improvement
- Each step builds upon the previous one
- Conduct a real-world BIA and risk analysis
- Test and validate your DR plan
- Apply templates to your use case
- Gain practical, hands-on experience

Course Outline and Structure

- Step-by-step lifecycle approach
- Lays the groundwork with foundational knowledge
- Moves into the practical aspects of DR planning
- Templates included for every stage
- Ends with a real DR plan build
- CloudCore use case throughout
- Assignments simulate real DR planning
- Logical flow from theory to practice

Meet CloudCore Technologies Inc.

- SaaS company based in Austin
- Mid-sized team, remote + in-house
- Serves 10,000+ small business clients
- Offers 24/7 cloud productivity software
- Sensitive to outages and downtime
- Uses cloud and third-party services
- Templates pre-filled with CloudCore data
- Ensures DR planning works in practice, not just in theory



Tools and Templates Provided

- DR Plan Template
- Business Impact Analysis Template
- Risk Assessment Matrix
- Recovery Strategy Worksheet
- DR Testing Checklist
- Communication Plan Template
- Post-Incident Report Template
- All downloadable and editable



What Is Disaster Recovery?

Aspect	Description
Purpose	Restore systems, data, and services post-disruption
Focus	IT systems, data, applications, infrastructure
Triggered by	Natural, cyber, human, or technical disasters
Common Components	Backups, RTO/RPO, recovery teams, documentation
Output	Documented DR plan with clear procedures
Beneficiaries	All organizations, from startups to enterprises

DR vs Business Continuity

Feature	Business Continuity	Disaster Recovery
Scope	Whole organization	IT systems and infrastructure
Objective	Maintain operations	Restore lost or damaged systems
Covers	People, processes, facilities	Data, applications, servers
Example Activity	Relocate staff	Recover database from backup
Owned By	Senior Management / Business Units	IT / Security Teams
Timeline	During and after a disruption	Post-disruption

Types of Disasters

Category	Examples	DR Considerations
Natural	Floods, earthquakes, hurricanes	Data center redundancy, physical site planning
Technical	Server failure, power outage	Backup power, hardware failover systems
Human	Misconfiguration, accidental deletion	Training, access controls, audit logs
Cyber	Ransomware, DDoS, phishing	Detection tools, encryption, response plans

Why DR Matters to Every Organization

Benefit	Description
Minimizes Downtime	Enables fast recovery of systems and services
Reduces Financial Loss	Prevents revenue loss during outages
Improves Decision-Making	Provides step-by-step response procedures
Builds Trust and Confidence	Reassures clients, partners, and auditors
Supports Compliance	Aligns with legal, regulatory, and contractual obligations
Enhances Preparedness Culture	Makes risk management part of daily operations

Get Executive Buy-In and Define Objectives

- Secure commitment from leadership
- Link DR to business risks and goals
- Set measurable RTO and RPO targets
- Include testing and training goals
- Align DR with compliance and SLA needs
- CloudCore example: 4-hour RPO, 12-hour RTO
- Maintain DR readiness across key teams

Form the DR Team and Assign Roles

Role	Responsibility
DR Lead	Oversees plan creation and execution
System Recovery Owner	Restores servers, data, and infrastructure
Communications Officer	Manages internal and external communication
HR Liaison	Coordinates with staff and manages logistics
Executive Sponsor	Provides direction and escalates decisions

Set Scope and Boundaries of the DR Plan



- Define included systems and locations
- Focus on mission-critical services
- Clarify in-scope vs. out-of-scope items
- Keep initial scope realistic
- Include expected disaster scenarios
- Prioritize customer-impacting functions
- CloudCore: product systems in, labs out

Create and Approve the DR Team Charter

- Formalizes team structure and authority
- Lists roles and responsibilities
- Defines DR meeting frequency
- Sets communication and escalation paths
- Reviewed and signed by leadership
- Improves clarity and accountability

CloudCore: quarterly drills + CIO escalation

What Is a Business Impact Analysis?

- Identifies critical functions and systems
- Foundational to DR plan effectiveness
- Measures disruption impact over time
- Defines RTO and RPO for each process
- Used by CloudCore for SLA-critical systems
- Based on interviews and business data
- Combines business, IT, and finance input
- Supports informed recovery strategy decisions



Identify Critical Functions and Dependencies

Prioritize based on operational impact

List and rank key business functions

Ask: What if this stops working?

Look at internal operations such as finance

CloudCore example: support and billing

Use results to guide recovery sequence

Include people, vendors, infrastructure

Document in function–dependency matrix

Define RTO and RPO

System	RTO	RPO
SaaS Application	4 hours	15 minutes
Ticketing Platform	6 hours	30 minutes
Billing System	8 hours	1 hour
HR System	48 hours	24 hours



Assess Financial and Operational Impacts

- Identify financial, operational, reputational risks
- Estimate revenue loss and compliance fines
- Include downtime effects on staff and customers
- Reputational impacts include customer complaints
- Tie impact to real business metrics
- CloudCore: SaaS outage = $5K/hour loss
- Use scoring to rank recovery priority

Use the BIA Template (Walkthrough)

- Download and duplicate the BIA Template
- Fill in function names and owners
- Assign impact levels to each function
- Use to inform recovery strategy decisions
- Interview teams to gather inputs
- Review examples from CloudCore’s worksheet
- Document RTO, RPO, and dependencies
- Update regularly as systems evolve

Identify Threats and Vulnerabilities

- Identify realistic internal and external threats
- Document system and process vulnerabilities
- Focus on risks relevant to your operations
- CloudCore risks: code loss, cloud outages
- Vulnerabilities include lack of offline backups
- Use workshops or checklists to gather input
- Include cyber, physical, technical, and human risks
- Record findings in a risk register

Assess Risk Likelihood and Impact

Threat	Likelihood	Impact	Priority
DDoS Attack	Medium	High	High
Cloud Provider Outage	High	High	Critical
Insider Data Deletion	Medium	Medium	Medium
Fire at Office	Low	High	Low

Prioritize and Document Risks

- Prioritize and document threats in a risk register
- Use a structured format for clarity and tracking
- List risks with likelihood and impact ratings
- CloudCore example: SaaS attack prevention plan
- Assign priority and document mitigation steps
- Track implementation of controls over time
- Update register as systems or risks change
- Supports audits and compliance reporting



Apply Threat Modeling (Optional Advanced Step)

- Use threat modeling for complex environments
- STRIDE helps classify threat types
- Strengthened MFA and logging controls
- Map threats to system architecture
- Identify technical weaknesses and entry points
- CloudCore applied STRIDE to its APIs
- Not required, but adds depth to DR
- Supports secure recovery planning

Use the Risk Assessment Template

- Download the Risk Assessment Template
- List and rate each threat systematically
- Track actions and review regularly
- Assign controls and mitigation owners
- Assign a risk level and describe any mitigation
- Use CloudCore examples for guidance
- Makes risk data easy to visualize
- Supports planning, audits, and reporting

Map Systems to Recovery Priorities

System	Tier	RTO	Strategy Type
SaaS Platform	1	4 hours	Hot standby + failover
Customer Support Ticketing	1	6 hours	Cloud replication
CI/CD Pipeline (DevOps Tools)	2	24 hrs	Snapshot backup
HR Portal	3	48 hrs	Cold site recovery



Choose Recovery Approaches (IT Systems)

- Simpler ≠ worse—match method to function
- Use hot sites for high-speed recovery
- Warm/cold sites fit lower-tier systems
- Cloud replication ensures data availability
- Snapshots ideal for longer RTO systems
- CloudCore uses a hybrid approach
- Align tech strategy with business priority
- Choose based on risk, cost, and need

Address Non-IT Recovery Needs

- Plan for people and physical locations
- Ensure remote work and communications
- Identify and assess vendor dependencies
- Include HR, admin, and ops in planning
- CloudCore example: remote shift in COVID
- Document as part of recovery procedures
- Test non-IT response like IT systems
- Integrated with your IT recovery timelines

Document Strategy Decisions in the Worksheet

- Download the Recovery Strategy Worksheet
- Align systems to strategies and owners
- Integrates outputs from Steps 3–5
- Include RTO, RPO, and resource needs
- CloudCore example shows live updates
- Keep updated with system changes
- Core part of DR documentation
- Useful for audits and stakeholder reviews

Structure of the Disaster Recovery Plan

Section	Content Description
1. Introduction	Purpose, scope, plan owner, document control
2. Roles & Responsibilities	Recovery team structure, contacts, escalation path
3. Recovery Procedures	System-specific recovery instructions and sequences
4. Communication Plan	Stakeholder messages, timing, internal/external contacts
5. Alternate Site Procedures	Office relocation, remote work, equipment access
6. Vendor Coordination	Critical vendors, SLAs, emergency contacts
7. Appendices	BIA, risk register, strategy worksheet, test results

Write System Recovery Procedures



- One procedure per critical system
- Include prerequisites, tools, and credentials
- Write step-by-step recovery instructions
- Add screenshots or commands where useful
- Assign clear ownership for execution
- CloudCore uses verified restore scripts
- Review and test each procedure regularly
- Update after system or vendor changes

Build the Contact Directory and Escalation Paths

- Embed contacts into recovery procedures
- Create a centralized, role-based contact sheet
- Define time-based escalation triggers
- Match contacts to DR responsibilities

- Delays occur because the right people cannot be reached in time
- Include primary/secondary contact methods
- **CloudCore example:** color-coded directory
- Update quarterly or after team changes

Use the DR Plan Template

- Designed for clarity and audit-readiness
- Template is divided into logical sections
- Adapt it to your organization by removing or expanding sections
- Includes guidance, examples, and structure
- Contains checklist-ready recovery sections
- CloudCore uses it as their live DRP
- Stored in internal documentation system
- Download the editable DR Plan Template

Why Communication Is Critical in DR

- Enables coordination under pressure
- Ensures stakeholders stay informed
- Prevents delays and builds trust
- Required by some regulations (e.g., GDPR)
- Avoids duplicated or missed recovery steps
- CloudCore uses pre-written updates and scripts
- Builds trust internally and externally
- Reduces panic and confusion



Identify Stakeholders and Notification Triggers

Trigger Event	Stakeholder Notified	Notification Time	Channel
App goes offline >15 mins	IT, DR Team, CIO	Immediate	SMS/Email
Customer login issues	Customer Support, Comms Team	< 30 mins	Slack/Email
Data breach suspected	Legal, CEO, Regulatory Body	< 1 hour	Secure Email
Recovery complete	All Stakeholders	< 15 mins after fix	Email/Portal

Prepare Pre-Written Messages and Templates

- Save time and reduce errors
- Prepare templates for major scenarios
- Include internal and external variations
- Keep language clear and calm
- Avoid speculation—share only confirmed info
- Prepare templates for incident acknowledgment
- CloudCore example: login outage response
- Customize templates by audience



Define Communication Roles and Channels

- Assign a central Communications Lead
- Designate approved external spokespersons
- Define support roles (IT, legal, support)
- Use Slack, Teams, email, SMS internally
- Use status pages, banners, or emails externally
- Match channels to audience needs
- CloudCore uses in-app + email alerts
- Keep internal and public info consistent

Use the Communication Plan Template

- Download and customize the template
- List stakeholders and communication triggers
- Assign owners and timing for each message
- Link pre-approved message drafts
- Log communication tests and outcomes
- CloudCore updates it quarterly
- Use during drills and live events
- Store alongside the full DR plan

Why Testing Is Essential

- Reduces errors during real disruptions
- Validates procedures, tools, and readiness
- Identifies outdated or missing elements
- DR without testing = risk exposure
- Builds staff confidence and coordination
- Supports audits, compliance, and certification
- CloudCore tests quarterly using real data
- Lessons from tests improve the plan

Types of DR Tests

Test Type	Description	Complexity	Impact
Checklist	Verify readiness without action	Low	Minimal
Tabletop	Walk through scenario with no system change	Medium	Low
Simulation	Restore data/systems in test environment	High	Medium
Full Interruption	Shutdown real systems and execute recovery	Very High	High

Define Test Objectives and Success Criteria

- Define what the test should prove
- Examples: RTO validation, comms timing, staff readiness
- Set measurable benchmarks for success
- Restore SaaS platform within the 4-hour RTO
- Notify stakeholders within the expected time frame
- Track deviations and improvement areas
- CloudCore example: restore + notify in 2 hours
- Use post-test report for audit trail



Conduct the Test and Record Findings

- Assign a facilitator to run the test
- Use real roles and documented steps
- Observe what works and what doesn’t
- Record delays, confusion, or errors
- Keep detailed notes on observations
- CloudCore logs test scenarios + outcomes
- Meet post-test for team debrief
- Update DR docs based on results

Use the DR Testing Checklist

- Download the DR Testing Checklist
- Adapt for tabletop, simulation, or full tests
- Covers pre-test, execution, and review steps
- CloudCore uses it for all DR drills
- Supports audit, ISO, and internal reviews
- Tracks lessons learned and next steps
- Use as part of DR documentation
- Improves consistency across tests





Activate the Disaster Recovery Plan

- Declare DR incident when trigger is met
- DR Lead confirms, then activates the plan
- Notify teams and begin recovery execution
- Follow documented priorities and workflows
- CloudCore: auto-alert → DR plan activation
- Avoid ad hoc decisions during activation
- Minimize confusion and downtime
- Document every step for review

Manage the First 24 Hours

- Stabilize, communicate, and start recovery
- Confirm availability of DR team members
- Log every action and decision
- Use central command center or dashboard
- Begin restoring Tier 1 systems
- CloudCore: 30-minute updates and dashboard sync
- Set expectations with stakeholders early
- Use pre-written communications

Conduct a Post-Incident Review

- Review what went well and what didn’t
- Include full DR team and leaders
- Build a timeline of actions and recovery
- CloudCore holds review within 5 days
- Document gaps and assign action items
- Update DR plan and templates afterward
- Show audit trail of lessons learned
- Support continuous improvement mindset

Complete the Post-Incident Report



- Summarize incident details and recovery phases
- Include timelines, team actions, and outcomes
- Record communication and escalation points
- Review if RTOs/RPOs were met or missed
- CloudCore: standardized template + action log
- Store securely with DR documentation
- Supports audits, leadership reporting, and training
- Use real data to drive DR updates

Use the Post-Incident Report Template

- Download the Post-Incident Report Template
- Includes sections for executive summary
- Add lessons learned and improvement tasks
- Helps document gaps and progress
- Drives actionable updates to the DR plan
- Record timelines, system status, and comms
- Include CloudCore sample as a reference
- Use after real events or major drills

Overview of the Final Project

- Apply what you've learned step by step
- Build a complete and usable DR plan
- Create a real deliverable for your portfolio
- Use your organization or CloudCore example
- Compile templates from previous assignments
- Each section builds on previous assignments
- Reference CloudCore's completed DRP
- Use this as benchmark to your own work

Components to Include in Your DR Plan

- Executive summary, scope, and goals
- DR team roles and contact details
- Completed BIA and risk assessment
- Recovery strategy worksheet
- System recovery procedures
- Communication and escalation plan
- Testing calendar and update schedule
- Appendices with all filled templates



Assignment Instructions and Submission

Use the Final Project Checklist

Follow the DRP template or build your own

CloudCore sample is available for reference

Anonymize real data if using your workplace

Export as PDF or Word format

Submit if required or use internally

Celebrate completing your own DR plan!

Review it with your DR team

Keep the DR Plan Updated

- DR plans must evolve with the business
- Review annually or after major changes
- Use version control to track updates
- Share changes with the full recovery team
- CloudCore updates quarterly across teams
- Assign review tasks by section or role
- Updates are logged and versioned
- Present to leadership as part of regular risk



Maintain DR Readiness Through Practice

- Integrate DR duties into team routines
- Rehearse roles through regular tabletop drills
- Train new team members during onboarding
- Keep all departments DR-aware
- CloudCore simulates stakeholder comms quarterly
- Use incident simulations as learning events
- Test backups and procedures frequently
- Practice builds calm and competence

Monitor Metrics and Track Improvements

- Track recovery performance against targets
- Review missed RTO/RPO vs planned levels
- Measure test participation and procedure success
- Monitor open issues from past incidents
- Use dashboards to share results with leaders
- Highlight areas of improvement
- Metrics support audits and investment decisions
- CloudCore reports 4 DR KPIs quarterly

Build a Culture of Resilience

- Make DR part of organizational values
- Leaders must model and support readiness
- Reward DR participation and improvement
- Encourage openness about risks and gaps
- Combine structure with adaptability
- Build teams that expect and overcome disruption
- CloudCore treats DR as a company habit
- Discuss recovery KPIs like any business metric



Course Conclusion – Your Path to Resilient Recovery

- You now have a complete DR planning framework
- Use your filled templates as living documents
- Learn from disruptions and apply improvements
- Update your plan regularly and test often
- Train your team and assign clear roles
- Share your plan with leadership and auditors
- Build resilience into company culture
- Be ready—not just compliant—when disruption comes